

JoTouch Boot Controller Training Results (generalized FMG -> commands)

- Purpose: generalized sensor->actuator relationship for the Teensy boot controller, until per-patient calibration replaces the weights
- Pipeline: AdamW + OneCycleLR + gradient clipping + early stopping (patience 8), up to 40 epochs, batch 64
- Split: 70% train / 15% val / 15% untouched test

Boot model	Output (target)	Epochs	Test MAE	Test RMSE	Test R ²	Latency (ms)	Weights
zenodo	stage position (X, Y) normalized - 2-axis continuous control	40	0.0195	0.0307	0.917	1.43	jotouch_boot_zenodo.pt
plos	grasped load [0..1] - grip closure intensity	17	0.3052	0.3524	0.024	1.61	jotouch_boot_plos.pt

Control chain on Teensy

1. Stage 1 (these weights): FMG window [8 ch x 20] -> continuous command(s).
2. Stage 2 (synergy table, hand-designed): $\text{motor_cmd}[7] = \text{intensity} * \text{Posture_g}[7]$.
3. Stage 3 (mechanical): 7 motors drive 12-16 finger DOFs via tendons/linkages.

Input contract: 8 FMG channels x 20 timesteps @100 Hz, min-max normalized per channel. Normalization stats MUST come from the device calibration step on the patient's own band (open-dataset stats in Content/open_fmg_data/*_norm_stats.npz are reference only).